MIL-R-81553A(AS)
30 August 1983
SUPERSEDING
MIL-R-81553(AS)
9 April 1968

MILITARY SPECIFICATION

REGULATOR, CHEST MOUNTED, 100 PERCENT OXYGEN, POSITIVE PRESSURE, CRU-79/P

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 100 percent oxygen, positive pressure, chest mounted regulator, type CRU-79/P.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-T-60	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-22191	Barrier Material, Transparent, Flexible, Heat Sealable
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33658	Fitting End, Hose Connections, Standard Dimensions for

HANDBOOKS

MILITARY

MIL-HDBK-695	Rubber Products: Recommended Shelf Life

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

62A-116-C48	Piezometer
62A-116-C49	Tube Piezometer
62A-116-C50	Ring Piezometer
62A-116-C51	Union Piezometer

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition function should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM D 1149-78A	Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The regulators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified (see 6.2.1c), a sample shall be subjected to first article inspection (see 4.4 and 6.4).

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.4.2.1 Age. Elastomer components except silicone shall not be more than 12 months old from the date of manufacture to the date of delivery to any Government service or to any airframe or accessory manufacturer.

3.4.2.2 Elastomer components. The elastomer components shall be composed of an ozone resistant composition, which shall not bloom, and shall meet the specified ozone resistance performance requirements.

3.4.2.3 Storage life. Elastomeric materials selected shall have an expected shelf storage life of 8 years or more based on the guidelines of MIL-HDBK-695 when such materials will provide the required physical, mechanical and chemical properties.

3.4.3 Protective treatment. When materials are used in the construction of the unit that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip or scale during normal service life or under extremes of environmental conditions shall not be used.

3.5 Standard parts. MS and AN Standard parts shall be used where they suit the purpose.

3.6 Design and construction. The chest mounted oxygen regulator shall be designed in accordance with figure 1. The regulator size shall not exceed the specified envelope dimensions.

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3.6.1 Oxygen inlet. A filter capable of filtering materials approximately 20 microns in size shall be provided in the oxygen inlet to prevent foreign particles from entering the regulator while in service. The filter shall be so installed that it can be easily removed or replaced.

3.6.2 Moisture. The design shall include provisions for preventing moisture from collecting within the regulator.

3.6.3 Plugs. All openings in the regulator shall be closed with caps or plugs to prevent foreign matter from entering the regulator during shipping and storage.

3.7 Performance.

3.7.1 Outlet pressures. The outlet pressures of the regulator, when tested as specified in 4.8.2, shall be in accordance with the requirements of table I. The maximum outlet pressure of the regulator shall not exceed 20 inches of water.

3.7.2 Noise level control. The 1/3 octave band analysis of the regulator, after test specified in 4.8.3, shall not exceed the decibel levels (db) of table II.

3.7.3 Leakage.

3.7.3.1 Body leakage. The regulator, when tested as specified in 4.8.4.1, shall not show any evidence of leakage through the body of the regulator.

3.7.3.2 Demand valve leakage - outward. The outlet pressures of the regulator, when tested as specified in 4.8.4.2, shall be in accordance with values specified in table I. The outlet pressures after the five-minute period shall be the same as the initial outlet pressures.

3.7.4 Overload.

3.7.4.1 Outlet. The regulator, when tested as specified in 4.8.5.1, shall not show any evidence of leakage or damage.

3.7.4.2 Inlet. The regulator, when tested as specified in 4.8.5.2 under zero flow conditions, the outlet pressure shall be not greater than 2.5 inches of water.

3.7.5 Extreme temperature.

3.7.5.1 High temperature. The regulator, when tested as specified in 4.8.6.1, shall show no evidence of mechanical or material failure and the outlet pressure shall be within plus or minus 0.5 inch of water of the outlet pressure requirements specified in table I.

3.7.5.2 Low temperature. The regulator, when tested as specified in 4.8.6.2, shall show no evidence of mechanical or material

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failure and the outlet pressure shall be within plus or minus 0.5 inch of water of the outlet pressure requirements specified in table I.

3.7.6 Vibration. The regulator, when tested as specified in 4.8.7, shall show no evidence of mechanical or material failure and shall operate in accordance with table I.

3.7.7 Acceleration. The regulator, when tested as specified in 4.8.8, shall not show any change in normal flow and shall show no evidence of mechanical or material failure.

3.7.8 Endurance. The regulator, when tested as specified in 4.8.9, shall show no evidence of mechanical or material failure.

3.7.9 Repeat of noise level control test. The 1/3 octave band analysis of the regulator, after tests specified in 4.8.10, shall not exceed the decibel levels (db) of table II.

3.7.10 Underwater breathing capabilities. The regulator, when tested as specified in 4.8.11, shall be capable of delivering an adequate supply of oxygen and the subject shall be capable of breathing from the regulator while in the submerged condition.

3.7.11 Ozone resistance. The elastomer test slabs, when tested as specified in 4.8.12, shall not show any evidence of checking, cracking, or other damage.

3.8 Interchangeability. All parts, subassemblies and assemblies having the same part number shall be dimensionally and mechanically interchangeable.

3.9 Weight. The weight of the oxygen regulator shall not exceed 5 ounces.

3.10 Color. The color of the regulator shall be black, approximately matching color number 37038 of FED-STD-595.

3.11 Finish. Aluminum and aluminum alloy parts shall be protected in accordance with MIL-A-8625 or MIL-C-5541.

3.12 Identification of product. The regulator shall be identified in accordance with MIL-STD-130, except that the Federal Stock Number shall be omitted from the nameplate.

3.12.1 Serial numbers. The regulators shall be identified by individual serial numbers which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive numbers to cover the entire acquisition quantity (see 4.6.2).

3.13 Cleanliness. The regulator shall be free of grease and other organic materials.

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3.14 Workmanship. The regulator shall be uniform in quality and shall be free from irregularities, defects or foreign material which could adversely affect safety, performance, reliability, or durability. The interior of the regulator shall be free from odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Quality conformance verification inspection (see 4.6).

4.3 Qualification inspection. Qualification inspection shall be as specified in table III.

4.3.1 Qualification samples. Qualification samples shall consist of four regulators, three elastomer test slabs, 6 by 6 by 0.075 inch, composed of the identical ozone-resistant composition and cure that was used in manufacturing the elastomer components and two sets of manufacturer's drawings for the samples submitted (see 6.3.1). Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-R-81553A(AS) and number under authorization (reference authorizing letter and number) (see 6.3).

4.3.2 Retention. The retention of qualification shall consist of periodic verification to determine compliance of the qualified regulator with the requirements of this specification. The time and

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method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article.

4.4.1 First article inspection. First article inspection shall be as specified in table IV (listed sequence mandatory).

4.4.2 First article samples. Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit four complete regulators and three elastomer test slabs, 6 by 6 by 0.075 inch, composed of the identical ozone-resistant composition and cure that was used in manufacturing the elastomer components. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on a new contract may be waived at the discretion of the acquiring activity (see 6.2.1c). Approval of the first article samples or the waiving of the first article inspection does not exclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1d).

4.4.2.1 First article information. Upon completion of the first article inspection program, pertinent comments and recommendations will be forwarded by the Government activity responsible for the inspection program (see 6.2.1d) to the contracting officer. Disposition of samples will be as follows:

a. One approved sample will be initially returned to the contractor for use in monitoring production.

b. One approved sample will be retained by the test facility for reference during quality conformance verification inspection and will be returned to the contractor with the samples from the final production lot.

c. The remaining samples will be consumed or destroyed during the inspection program and shall not be considered as part of the quantity to be delivered under the acquisition document.

4.5 Quality conformance inspection. The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105. Quality conformance inspection shall be as specified in table V (listed sequence mandatory).

4.5.1 Sampling.

4.5.1.1 Inspection lot.

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4.5.1.1.1 Regulator. An inspection lot size shall be expressed in units of one regulator made under essentially the same conditions and from the same materials and components. The sample unit shall be one regulator.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing regulators, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing regulators, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of the regulators. The sample size, acceptance criteria, test and examinations required for the regulators shall be as specified in table V (listed sequence mandatory).

4.6 Quality conformance verification.

4.6.1 Quality conformance verification inspection. The quality conformance verification inspection shall consist of the following tests and examinations (listed sequence mandatory):

Visual Examination
Outlet Pressures
Leakage
Overload

4.6.2 Quality conformance verification samples. Upon completion of the tests and examinations specified in table V, a random sample shall be selected from each lot in accordance with MIL-STD-105, Inspection Level S-3. The sample size shall be based only on the applicable sample size code letter corresponding to the Inspection Level S-3. Each regulator, selected as a sample unit, shall be identified by its assigned serial number (see 3.12.1) and shall be forwarded to the inspection facility specified in the acquisition document (see 6.2.1g). Additionally, the serial numbers of the units within the lot, represented by the sample units, shall be furnished to the inspection facility.

4.6.3 Quality conformance verification approval. Upon completion of the quality conformance verification inspection, the Government activity responsible for conducting the inspection program (see 6.2.1g) shall report the results of tests, with recommendations, to the contract administration office specified in the acquisition document. Final acceptance of the lot from which the samples were selected shall be based upon completion of the inspection program by the Government Quality Assurance Representative/Specialist applying the applicable acceptance criteria specified in table V. Upon completion of this inspection program the samples will be returned to the contractor, at the contractor's expense.

4.7 Test conditions.

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4.7.1 Gas. Unless otherwise specified, the gas used in testing the regulator shall be oil-free nitrogen, 99.95 percent pure, in accordance with BB-N-411, Type I, Grade A, Class 1 or oxygen in accordance with MIL-O-27210, Type I.

4.7.2 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. Temperature and pressure shall be recorded at the time of inspection and, when required, the test results shall be corrected to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury (101.3 kPa) and 70°F (21.1°C). In the case of pressure drop measurements, the flow setting for conducting the test shall be corrected to NTP conditions before the test is conducted.

4.7.3 Adjustment of inlet supply pressure. At the start of the operational tests, the initial static test inlet pressure of 40, 50, 70 or 120 psig applies only to zero flow conditions. When the pressure reducing regulator device is not in the same ambient conditions as the oxygen breathing regulator, means shall be provided to determine the above noted pressure at the ambient conditions of the breathing regulator. When flow is started, the inlet supply pressure shall not be allowed to drop more than 8 psig at 100 lpm ambient flow for inlet pressures of 50, 70 or 120 psig. Under no condition shall the inlet supply pressure be allowed to drop below 40 psig.

4.7.4 Orientation. The regulator shall operate as specified herein in any position which it is mounted.

4.7.5 Piezometer assembly. Unless otherwise specified, the regulator shall be operated with a test gas (see 4.7.1) at an inlet supply pressure of 40 to 120 psig. The outlet pressure shall be measured using a piezometer assembly conforming to NAVAIR drawings 62A-116-C48, 62A-116-C49, 62A-116-C50 and 62A-116-C51. The piezometer shall be connected to the mask end of the regulator at a distance of not more than 1.0 inch from the regulator fitting. The regulator outlet pressures shall be measured at the piezometer union shown in NAVAIR drawing 62A-116-C48.

4.7.6 Vibratory control. If during any of the tests, the regulator vibrates, flutters, chatters, or makes noises, the regulator shall be rejected. A vibration or flutter is that condition in the regulator that causes the output flow and meter to fluctuate. Flutter dampers shall not be used to control or eliminate this condition. Flutter or vibration shall also be determined by using live subjects at various altitude conditions and shall take precedence over other types of tests when determining if flutter or vibration exists. During the subjective tests, the regulator shall be connected to an oxygen hose conforming to Scott Aviation part number 834-18, or equivalent, and shall be installed on an MBU-14/P oxygen mask.

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4.7.7 Test equipment.

4.7.7.1 Test stand. All flow and pressure tests should be performed on the Firewel Test Stand, P/N F053C007. Regulators, when tested on other test stands, shall be capable of fulfilling all flow and pressure test requirements of this specification when tested on the Firewel Test Stand.

4.7.7.2 Noise level control test set-up. Noise level control test set-up shall be in accordance with figure 2 and the test shall be conducted in accordance with 4.8.3. The set-up shall consist of Bruel & Kjaer test equipment, or equivalent, and shall include the following:

Description	Model No.
Digital Frequency Analyzer	2131
Condenser Microphone 1" Free Field	4145
Condenser Microphone Preamplifier	2619S
Graphic Level Recorder, With Voltage Control Paper Drive	2307A
Operation 2120 and 1614-10 Synchronous or Charge Cable	AQ-0035

4.8 Inspection methods.

4.8.1 Visual examinations.

4.8.1.1 Regulator. Every regulator shall be examined visually for critical defects to determine conformance to this specification and applicable drawings. Every regulator, selected as a sample unit from the lot, shall be visually examined for minor defects to determine conformance to this specification. The classification of defects, table VI, shall be used to classify the defects found.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing regulators, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing and marking conform to this specification. The list of defects, table VII, shall be used to enumerate the defects found.

4.8.2 Outlet pressure. The regulator shall be connected to a pressure source capable of providing the inlet pressures indicated in table I. The regulator shall then be subjected to the pressures and conditions specified in 4.8.2.1 or 4.8.2.2, as applicable. The outlet pressure shall be measured as specified in 4.7.5, with the use of a piezometer assembly, and shall pass the requirements specified in 3.7.1.

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4.8.2.1 Outlet pressure - qualification and first article inspections. Qualification and first article inspections shall be performed utilizing the altitudes, inlet supply pressures and flows delineated in table I.

4.8.2.2 Outlet pressure - quality conformance and quality conformance verification inspections. Quality conformance and quality conformance verification inspections shall be performed in accordance with table I except that the tests shall be conducted utilizing ambient flows of 0, 50, and 100 lpm with inlet pressures of 50 and 120 psig at altitudes of ground level, 10,000 and 34,000 feet. In the pressure breathing range, tests shall be conducted utilizing ambient flows of 0 and 100 lpm with inlet pressures of 40 and 120 psig at altitudes of 35,000, 40,000 45,000 and 50,000 feet.

4.8.3 Noise level control. The regulator shall be installed as shown in figure 2. The regulator shall then be subjected to an ambient flow of 100 lpm at an inlet pressure of 70 and 120 psig. The noise level shall be measured and shall pass the requirements specified in 3.7.2.

4.8.4 Leakage.

4.8.4.1 Body leakage. The inlet and all other orifices of the regulator shall be capped and a pressure of 10 inches of water shall be applied to the outlet of the regulator for a period of 3 minutes. The regulator shall pass the requirements specified in 3.7.3.1.

4.8.4.1.1 Adjustment for body leakage. If the regulator incorporates a designed bleed flow that vents through the body, an adjustment to the body leakage requirement will be allowed. Any flow over the designed bleed flow will be considered leakage for the body leakage test.

4.8.4.2 Demand valve leakage - outward. The inlet of the regulator shall be subjected to a pressure of 40 to 120 psig while maintaining zero flow conditions at ground level. After a five-minute period, the outlet pressure of the regulator shall be measured and shall pass the requirements specified in 3.7.3.2.

4.8.5 Overload.

4.8.5.1 Outlet. The inlet and all other orifices of the regulator shall be capped and a pressure of 25 inches of water shall be applied to the outlet of the regulator and maintained for 2 minutes. The regulator shall pass the requirements specified in 3.7.4.1.

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4.8.5.2 Inlet. While maintaining a zero flow condition at ground level, a pressure of 150 psig shall be applied to the inlet of the regulator and shall be maintained for a minimum of 3 minutes. The regulator shall pass the requirements specified in 3.7.4.2.

4.8.5.3 Outlet pressure check. Upon completion of the tests specified in 4.8.5.1 and 4.8.5.2, the regulator shall be subjected to and pass the outlet pressure test, 4.8.2.

4.8.6 Extreme temperatures.

4.8.6.1 High temperature. The regulator shall be subjected to the high temperature test in accordance with MIL-STD-810, method 501.1, procedure II, except that the conditioning period shall be three hours at a temperature of 160 ± 5°F. After conditioning and while still at this temperature, the regulator shall be subjected to the outlet pressure requirements of table I and shall pass the requirements specified in 3.7.5.1. Upon completion of this test procedure, the regulator shall be returned to room temperature and shall then be subjected to and pass the outlet pressure and leakage tests, 4.8.2 and 4.8.4.

4.8.6.2 Low temperature. The regulator shall be subjected to the low temperature test in accordance with MIL-STD-810, method 502.1, procedure 1, except that the conditioning period shall be three hours at a temperature of -65 ± 5°F. After conditioning and while still at this temperature, the regulator shall be subjected to the outlet pressure requirements of table I and shall pass the requirements specified in 3.7.5.2. Upon completion of this test procedure, the regulator shall be returned to room temperature and shall then be subjected to and pass the outlet pressure and leakage tests, 4.8.2 and 4.8.4.

4.8.7 Vibration. The regulator shall be capped with end caps containing a minute bleed hole to relieve the buildup of pressure. A pressure of 80 ± 10 psig shall be applied to the inlet of the regulator. The regulator shall be vibrated for 3 hours at a frequency from 500 to 2500 cycles per minute, and a double amplitude of 0.018 to 0.020 inch. The regulator shall pass the requirements specified in 3.7.6. After vibration, the regulator shall be subjected to and pass the outlet pressure and leakage tests, 4.8.2 and 4.8.4.

4.8.8 Acceleration. A pressure between 50 and 120 psig shall be applied to the inlet of the regulator and an acceleration force of 7g shall be applied to the regulator in six different directions. One force shall be applied at one time, and the forces shall be in the directions shown in figure 3. During the application of each acceleration force, the regulator shall pass the requirements specified in 3.7.7.

4.8.9 Endurance. The regulator shall be subjected to 100,000 breathing cycles of endurance testing at 15 to 18 cycles per minute, and shall pass the requirements specified in 3.7.8. The peak

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delivery rate shall be 30 lpm (stp) for 90 percent of the time and 90 lpm (stp) for 10 percent of the time. The regulator shall then be subjected to and pass the outlet pressure test, 4.8.2.

4.8.10 Repeat of noise level control test. At the completion of the tests described in 4.8.6 through 4.8.9 the regulator shall be subjected to the noise level control test and shall pass the requirements specified in 3.7.9.

4.8.11 Underwater breathing capabilities. The regulator shall be connected to a hose conforming to Scott Aviation part number 834-18, or equivalent and the hose shall be connected to an MBU-14/P oxygen mask and fitted to a subject. The regulator shall be positioned 2 inches to 4 inches below the subject's suprasternal notch. The inlet oxygen pressure shall be maintained at 50 psig. The subject shall be submerged under 16 feet of water, for a minimum of 3 minutes, in both the upright and inverted positions. Under these conditions, the regulator shall pass the requirements specified in 3.7.10.

4.8.12 Ozone resistance (elastomer components). The three elastomer test slabs submitted for qualification or first article inspection shall be tested for ozone resistance. The test apparatus shall be in accordance with ASTM D 1149-78A. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to 100 $\pm$ 2°F and to give an exposure of ozone concentration of 120 $\pm$ 10 parts by volume of ozone per million parts by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall be examined under a 10X magnification and pass the requirements specified in 3.7.11.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2.1e).

5.1.1 Level A. Each breathing regulator shall be preserved in accordance with MIL-P-116, method IC-4. No contact preservation is required. Each breathing regulator shall be sealed within a snug fitting bag of 0.0015 inch gage material conforming to MIL-B-22191, Type III. The sealed breathing regulator shall be cushioned and packaged within a waterproof fiber can with sealed metal ends. The breathing regulator shall be cushioned in such a manner as to prevent damage to itself. Ten unit packages shall be packaged within a fiberboard container conforming to PPP-B-636, Type I, Class 2, Grade 3. The manufacturer's seam shall be adhesive sealed. All seams and joints shall be sealed with tape conforming to PPP-T-60.

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5.1.2 Level C. The breathing regulator shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1e).

5.2.1 Level A. One hundred breathing regulators, packaged as specified in 5.1.1, shall be packed in overseas type exterior containers conforming to PPP-B-601, PPP-B-621, PPP-B-591, PPP-B-636, or PPP-B-640, Class 2. Packages for Southeast Asia shall be packed in PPP-B-621 boxes, constructed from wood which has been sprayed with a solution of 2 percent copper naphthenate in mineral spirits. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B. One hundred breathing regulators, packaged as specified in 5.1.1 shall be packed in domestic type exterior containers conforming to PPP-B-601, PPP-B-621, PPP-B-591, PPP-B-636, or PPP-B-640, Class 1. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.3 Level C. The packaged breathing regulators, which the carrier requires to be packed, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. Nomenclature tags shall be inserted in each bag showing title, part number or other data as required by acquiring activity. In addition, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN REGULATORS."

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6. NOTES

6.1 Intended use. The regulator covered by this specification is intended for use in supplying breathing oxygen to aircrewmen in the operation of an aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Applicable QPL, Government designation and manufacturer designation required.

c. Whether first article inspection is required (see 3.2 and 4.4.2).

d. Name and address of the first article inspection facility and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.2 and 4.4.2.1).

e. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

f. Applicable methods of cleaning and preservation.

g. Name and address of the quality conformance verification inspection facility and the name and address of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.6 and 6.5).

h. Items of data required (see 6.2.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4.2.1	First article inspection reports	DI-T-5329 - Inspection test reports

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors are called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20360; however, authorization for qualification of products shall be obtained from the Commander, Naval Air Development Center, (Code 6031), Warminster, PA 18974.

6.3.1 Drawings. When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6 (see 4.3.1).

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.5 Laboratory information. The successful bidder will be furnished with the name of the quality conformance verification inspection laboratory and the Government activity responsible for conducting the inspection program at the time of award. The costs of the tests and examinations on samples initially submitted from a lot, as required by 4.6 shall be borne by the Government whether the tests and examinations are conducted by a Government laboratory or by a laboratory selected by the cognizant Government activity. Samples from a rejected lot shall not be resubmitted for tests and examinations, as required by 4.6, without the approval of the contracting officer. The costs of the tests and examinations on samples resubmitted from a reworked lot or from a new lot which is necessitated by the rejection of a previous lot shall be borne by the manufacturer whether the tests and examinations are conducted by a Government laboratory or by a laboratory selected by the cognizant Government activity.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project No. 1660-N463)

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TABLE I. Outlet pressures.

Altitude (feet)	Inlet Supply Pressure (PSIG)	Outlet Pressure @ 0.0 LPM Flow (In. of H_2O)	Outlet Pressure @ 50.0 LPM Flow (In. of H_2O)	Outlet Pressure @ 100 LPM Flow (In. of H_2O)
Ground level	40, 80, 120	0.5 to 2.5	0.5 to 2.5	1.0 to 2.5
10,000	40, 80, 120	0.5 to 2.5	0.5 to 2.5	1.0 to 2.5
34,000	40, 80, 120	0.5 to 2.5	0.5 to 2.5	1.0 to 2.5
35,000	40, 80, 120	0.5 to 3.0	0.5 to 3.0	1.0 to 3.0
40,000	40, 80, 120	5.0 to 9.5	5.0 to 9.5	5.0 to 9.5
45,000	40, 80, 120	10.5 to 15.5	10.5 to 15.5	10.5 to 15.5
50,000	40, 80, 120	14.0 to 20.0	14.0 to 20.0	14.0 to 20.0

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TABLE II. Noise level.

1/3 Octave Band Center Frequency	db (Referred To) 0.0002 dynes/cm^2
315	95 max
400	95 max
500	95 max
630	95 max
800	95 max
1000	95 max
1250	95 max
1600	95 max
2000	95 max
2500	95 max
3150	95 max
4000	95 max
5000	95 max
6300	95 max

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TABLE III. Qualification inspection.

Inspection	Requirement Paragraph	Paragraph Test Method
Visual examination	-	4.8.1
Outlet pressures	3.7.1	4.8.2
Noise level control	3.7.2	4.8.3
Body leakage	3.7.3.1	4.8.4.1
Demand valve leakage - outward	3.7.3.2	4.8.4.2
Overload	3.7.4	4.8.5
High temperature	3.7.5.1	4.8.6.1
Low temperature	3.7.5.2	4.8.6.2
Vibration	3.7.6	4.8.7
Acceleration	3.7.7	4.8.8
Endurance	3.7.8	4.8.9
Repeat of noise level control	3.7.9	4.8.10
Underwater breathing capabilities	3.7.10	4.8.11
Ozone resistance (elastomer components)	3.7.11	4.8.12

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TABLE IV. First article inspection.

Inspection	Requirement Paragraph	Test Method Paragraph
Visual examination	-	4.8.1
Outlet pressures	3.7.1	4.8.2
Body leakage	3.7.3.1	4.8.4.1
Demand valve leakage - outward	3.7.3.2	4.8.4.2
Overload	3.7.4	4.8.5
Underwater capabilities	3.7.10	4.8.11
Noise level control	3.7.2	4.8.3
Ozone resistance (elastomer components)	3.7.11	4.8.12

TABLE V. Sample size, acceptance criteria, tests and examination of the regulators.

Inspection	Method	Sample Size	Acceptance Criteria
Visual examination (See classification of defects)	4.8.1.1	Every regulator for critical defects. Inspection Level II for minor defects	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Outlet pressure	4.8.2	Every regulator	Reject all defective units.
Leakage	4.8.4	Every regulator	Reject all defective units.
Overload	4.8.5	Inspection Level S-2	An acceptable quality level of 1.5 defects per hundred units.
Noise level control	4.8.3	Every regulator	Reject all defective units.
Packaging	4.8.1.2	Inspection Level S-2	An acceptable quality level of 4.0 defects per hundred units.

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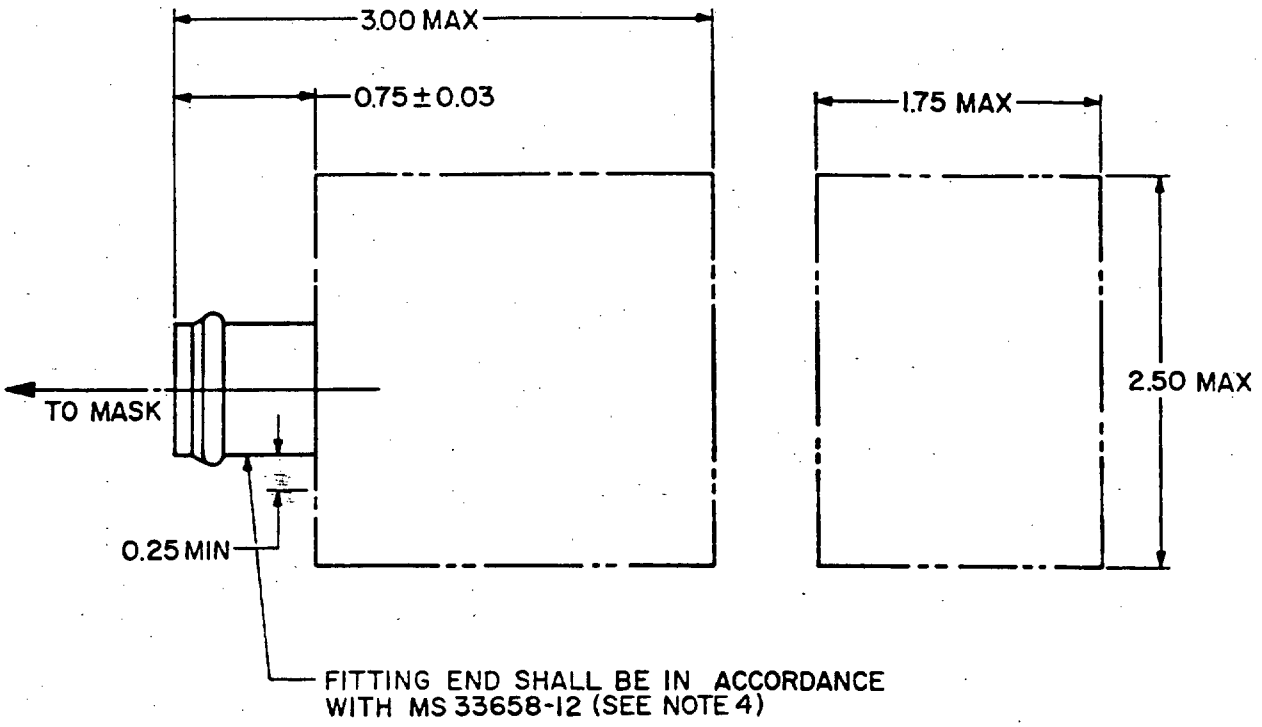
TABLE VI. Classification of defects for visual examination of the regulator.

Critical	Minor
1. Material imperfections - foreign matter embedded.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

TABLE VII. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, case liners, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

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DIMENSIONS IN INCHES

NOTES:

1. ANPT SHALL BE IN ACCORDANCE WITH MIL-P-7105.
2. REGULATOR SHALL INCLUDE A 1/8-27 ANPT INLET FITTING FOR 40-120 PSIG OXYGEN.
3. ENVELOPE DIMENSIONS INCLUDE ALL HOSE CONNECTING PARTS.
4. INTERNAL DIMENSION NOT APPLICABLE.

Figure 1. Chest mounted regulator (envelope dimensions).

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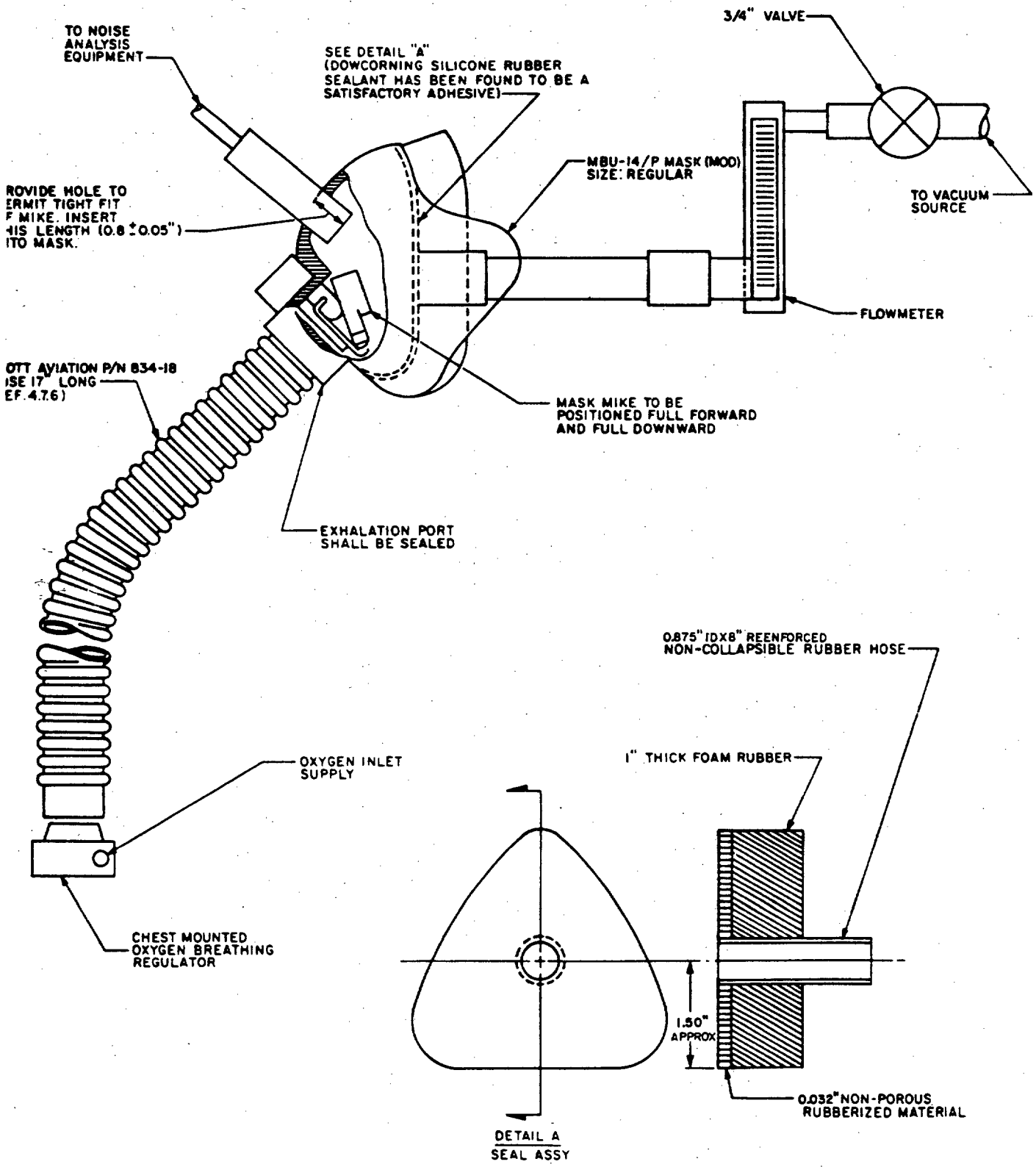
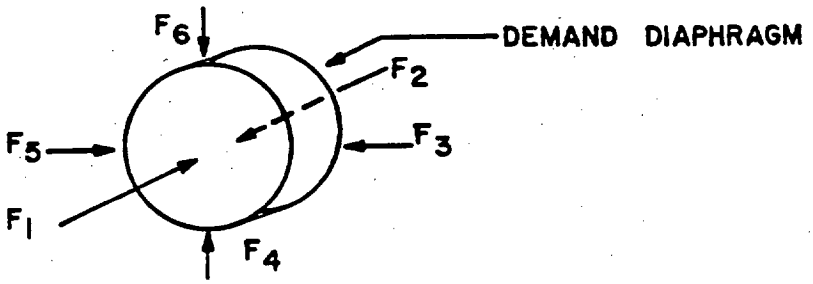


Figure 2. Noise level control test set-up.

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NOTE: F_1 AND F_2 ARE PERPENDICULAR TO THE PLANE OF THE DEMAND DIAPHRAGM. F_3, F_4, F_5 AND F_6 ARE PARALLEL TO THE PLANE OF THE DEMAND DIAPHRAGM.

Figure 3. Direction of applied forces.

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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